

CHICAGO JOLIET, IL, USA

WMO: 725345

Lat: 41.518N

Lon: 88.176W

Elev: 582

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 14834

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) 2.9	(d) -11.1	(e) 3.1	(f) -0.1	(g) -6.0	(h) 4.1	(i) 5.0	(j) 20.8	(k) 31.8	(l) 19.0	(m) 31.6	(n) 5.3	(o) 280	(p) 0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	91.4	75.5	89.6	74.6	87.0	73.1	79.2	87.7	77.2	85.6	75.4	83.0	7.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	143.5	84.3	74.7	132.5	81.8	72.9	124.6	80.2	43.1	87.9	40.9	85.7	39.2	82.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.4	17.1	15.5	-11.0	95.0	7.9	3.2	-16.7	97.2	-21.3	99.1	-25.7	100.9	-31.4	103.2		
			-11.5	82.3	7.4	1.9	-16.9	83.7	-21.2	84.8	-25.3	85.9	-30.7	87.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	24.7	27.0	39.0	50.5	60.8	70.4	74.0	72.2	65.8	52.8	41.3	30.6		
	DBStd	19.62	12.16	11.90	11.21	9.52	9.13	6.64	5.75	5.42	7.75	9.24	9.38	10.48		
	HDD50	2897	784	646	376	108	12	0	0	0	1	76	289	605		
	HDD65	6045	1248	1064	810	443	195	23	3	7	82	393	711	1066		
	CDD50	3224	1	3	34	123	347	611	746	689	474	164	28	4		
	CDD65	895	0	0	3	8	65	184	283	231	105	16	0	0		
	CDH74	8468	0	0	23	125	659	1770	2808	1993	947	142	1	0		
	CDH80	2858	0	0	2	21	200	617	1047	647	294	30	0	0		
(14) Wind	WSAvg	6.2	7.2	7.1	7.6	8.2	6.7	5.4	4.5	4.1	4.8	5.9	6.6	6.9		
(15) Precipitation	PrecAvg	35.8	1.8	1.6	2.2	3.6	4.2	4.2	3.9	3.9	3.1	2.9	2.6	1.9		
	PrecMax	45.4	4.1	5.6	5.0	6.9	7.4	9.9	7.7	9.6	12.6	7.9	5.9	4.9		
	PrecMin	23.7	0.4	0.4	0.7	1.5	0.7	0.9	1.2	1.6	0.6	0.5	0.4	0.5		
	PrecStd	5.0	1.0	1.1	1.1	1.5	1.6	2.2	1.5	1.6	2.4	1.9	1.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	62.8	77.9	82.3	90.0	93.6	96.6	93.1	92.2	84.2	70.3	62.9		
		MCWB	53.5	53.7	62.7	63.8	71.6	74.6	77.2	76.8	74.6	70.9	57.5	58.7		
	2%	DB	51.5	54.7	68.5	77.5	85.9	90.4	91.4	90.0	88.0	78.3	64.3	54.7		
		MCWB	48.1	48.9	57.7	60.6	68.4	73.7	77.0	75.6	72.9	64.7	55.9	51.1		
	5%	DB	45.6	47.9	62.1	73.0	81.2	87.6	89.2	87.3	82.7	73.0	60.9	49.6		
		MCWB	42.9	43.4	54.0	58.5	66.3	72.1	76.0	74.0	69.8	62.8	54.1	46.6		
	10%	DB	40.8	42.5	56.6	68.5	77.2	83.0	86.2	83.2	79.5	69.2	55.9	45.2		
		MCWB	37.9	38.5	49.5	56.9	65.3	70.6	73.7	71.5	67.5	61.0	49.6	42.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	57.5	64.1	68.3	75.4	80.1	82.3	81.4	77.9	73.3	62.8	59.0		
		MCDB	57.8	59.8	73.6	78.3	85.2	89.2	90.0	89.8	87.3	80.5	67.6	61.5		
	2%	WB	47.8	50.0	60.0	64.2	71.8	76.8	80.0	78.3	75.5	69.4	58.5	52.0		
		MCDB	50.5	54.5	68.1	73.3	80.3	85.6	88.4	86.1	83.9	73.1	62.5	54.0		
	5%	WB	42.3	44.1	55.9	61.0	69.5	74.5	77.5	76.0	72.9	65.4	54.5	46.3		
		MCDB	45.7	48.1	61.3	70.4	77.1	82.8	85.8	83.3	79.5	70.6	59.2	48.1		
	10%	WB	37.7	38.7	50.3	57.8	67.3	72.7	75.7	74.2	70.6	61.8	51.0	42.0		
		MCDB	40.5	42.1	55.5	67.4	74.6	80.6	83.3	80.7	77.1	67.7	55.8	44.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	16.1	17.7	20.9	21.4	20.7	19.9	19.6	21.4	19.7	16.8	13.2		
		MCDBR	18.8	22.5	24.7	28.1	25.9	23.9	22.2	22.7	24.2	25.8	23.2	18.0		
		MCWBR	15.8	17.4	16.8	16.0	13.0	11.2	10.8	11.3	11.4	14.9	16.3	15.6		
	5% WB	MCDBR	17.9	20.7	22.2	24.8	21.3	20.2	19.8	19.0	20.1	20.0	19.9	16.8		
		MCWBR	15.6	17.3	16.6	16.3	13.3	11.4	11.7	10.9	10.7	14.8	16.1	15.6		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.315	0.349	0.403	0.433	0.439	0.454	0.445	0.416	0.359	0.314	0.296		
	taud		2.424	2.372	2.373	2.233	2.203	2.230	2.184	2.201	2.252	2.412	2.488	2.447		
	Ebn at Noon		272	283	285	275	268	266	261	259	259	263	263	262		
	Edh at Noon		25	30	35	43	45	44	46	43	39	29	23	23		
(44) All-Sky Solar Radiation	RadAvg		524	803	1139	1474	1681	1918	1920	1688	1416	937	602	443		
	RadStd		52	107	86	120	139	135	104	96	121	104	57	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page